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ED-Elas2D: An Educational Program for Computer-aided Training in Structural Analysis Using the Finite Element Method

BENJAMÍN SUÁREZ, ELENA BLANCO, LLUÍS GIL, PERE ZAPATA & EUGENIO OÑATE

SUMMARY *In this paper, a methodology for computer-aided training of structural problems by the finite element method is presented. The resulting educational program, named ED-Elas2D, has been developed by a group of researchers at the School of Civil Engineering at the Technical University of Catalonia (UPC) in Barcelona, UPC. The program combines the standard concept which appears in the traditional teaching of finite element methods for structural analysis, such as the introduction to the basic structural theory, data definition (pre-processor), description of the analysis procedure (processor) and interpretation of results (post-processor). The program is addressed to both students who enrol in the study on finite element methods and to academic and technical staff in universities and companies who wish to update their knowledge of finite element techniques. The educational program ED-Elas2D works in the Windows system and it is characterized by its interactivity and graphic utilities.*

1. Introduction

The finite element method (FEM) is a standard technique for the solution of problems in science and technology with the help of the computer. Among the usual applications of FEM, we can note the analysis of solids and structures in different fields of civil, mechanic, aerospace and naval engineering, among others.

The principles of FEM are based in concepts very similar to those used in the analysis of structural systems, such as bar structures and electrical or pipeline networks using matrix analysis techniques. The education of the user in this field demands the knowledge of topics such as the discretization of the analysis domain in the assembly of the matrices and vectors of each element, the description of the boundary conditions, the application of the loading and the final solution of the global system of equations for obtaining the nodal unknowns and other parameters of interest for each element, such as displacements, strains and stresses in a structural problem.

The educational environment ED-Elas2D [1] tries to give a solution to all these problems while aiming, at the same time, to help the structural engineer in the understanding of FEM. The program follows the methodology proposed by the authors in the development of the educational programs ED-Beams and ED-Frames [2, 3], designed for the computer-aided training in linear elastic frames using matrix methods. Indeed, FEM uses matrix techniques to solve an equation system. This means that a

good knowledge in this field is essential for a better understanding of all the steps involved in finite element computations [4].

The topic chosen to initiate the study of FEM was the linear elastic analysis of structures under plain stress or plain strain conditions, i.e. structures satisfying the assumptions of 2D elasticity. Starting from the experience gained in the development of previous educational programs ED-Beams and ED-Frames, the authors have enhanced some aspects of the working environment, in particular those referring to concepts such as the pedagogical content and interactivity.

It is worth noting that although the problem which specifically solves ED-Elas2D is the analysis of 2D solids and structures, many of the concepts which appear in the program are completely general and useful to understand FEM solution of other problems in different fields of science engineering [5].

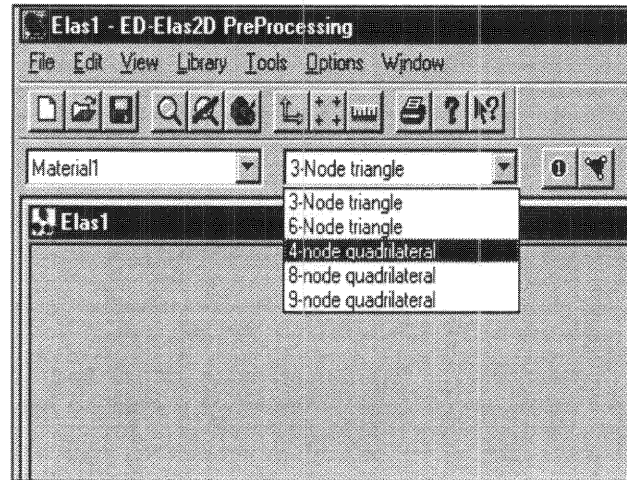


FIG. 1. Definition of a new problem. Element types accepted by the program.

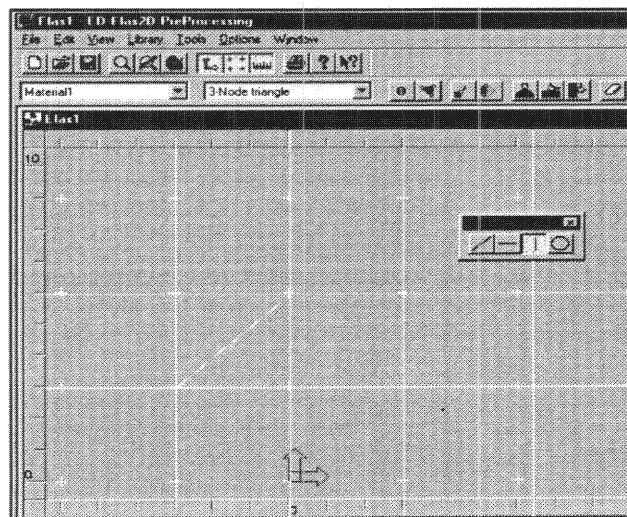


FIG. 2. Auxiliary lines, axis and frame for the generation of the problem.

2. Description of the Educational Environment

The FEM presents an intrinsic difficulty which affects the learning process. Although the description of the theoretical concepts is relatively easy using the classical educational techniques (i.e. blackboard, books, notes, etc.), the development of examples and exercises of application for some real structures is quite difficult using these tools. Generally, finite element teachers find that the blackboard is very limited for tackling the solution of practical structures which could show the power of FEM and they typically resort to computer workshops. For these reasons the students can only have access to simple academic examples involving one or two elements which are far from showing all the capabilities and difficulties of FEM.

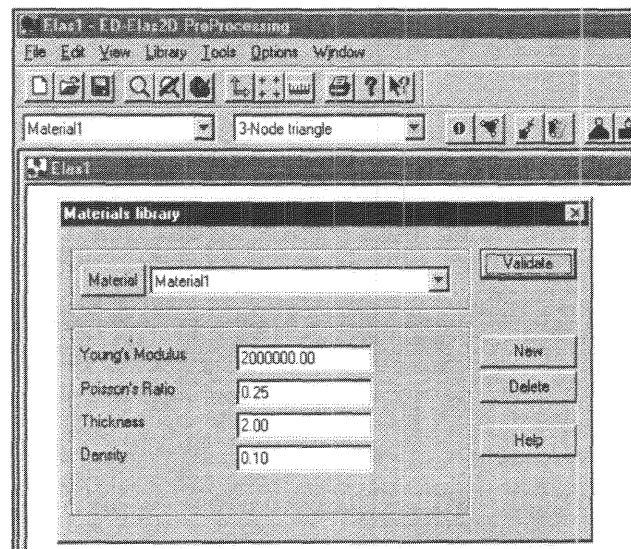


FIG. 3. Library of materials. Dialogue box for the definition of a new material.

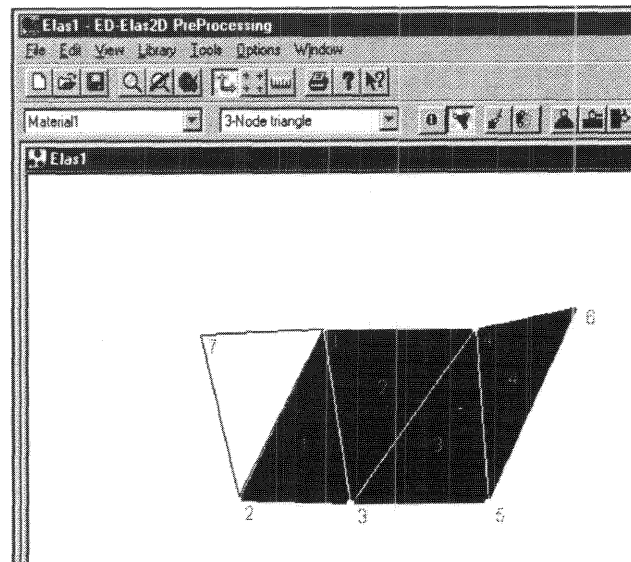


FIG. 4. Definition of the geometry and the mesh of a problem.

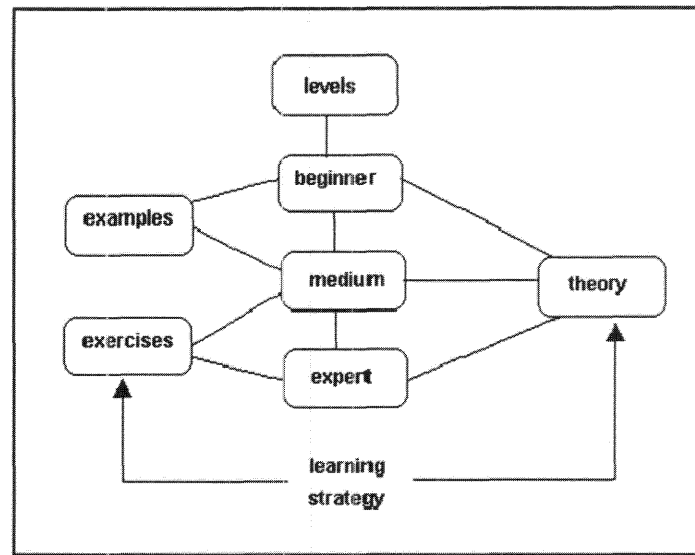


FIG. 5. General flow chart of the processing block of ED-Elas 2D.

FD-Elas2D Processing - Help						
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Contenido	Búsqueda	Avanz	Imprimir	<	>	
Index of theory						
<u>Introduction</u>						
<u>General setting out of the problem</u>						
<u>The continuous problem</u>						
Kinematic relations						
Constitutive relation						
Equilibrium equation						
<u>The discrete problem</u>						
Equation of equilibrium						
Displacement field						
Strain field						
Stress field						
Stiffness matrix						
<u>External forces</u>						
<u>Boundary conditions</u>						
<u>Results</u>						
Displacements						
Stresses						
Strains						
Cracks						

FIG. 6. List of theoretical lessons.

On the other hand, there are many FEM commercial programs which can be used as a base for computer tutorials. The main difficulty of these programs is the lack of transparency to be used as a fully teacher environment. In many cases, these commercial programs operate like a blackbox and therefore the student is practically limited to the introduction of the problem data and the interpretation of the results and has no access to the details of the analysis.

In a general sense, it can be said that the educational program, in addition to allowing the access to all the data definition and the analysis of results stages, should also have certain practise additional facilities in order to introduce the user adequately to a detailed knowledge of the method. Among these we can note:

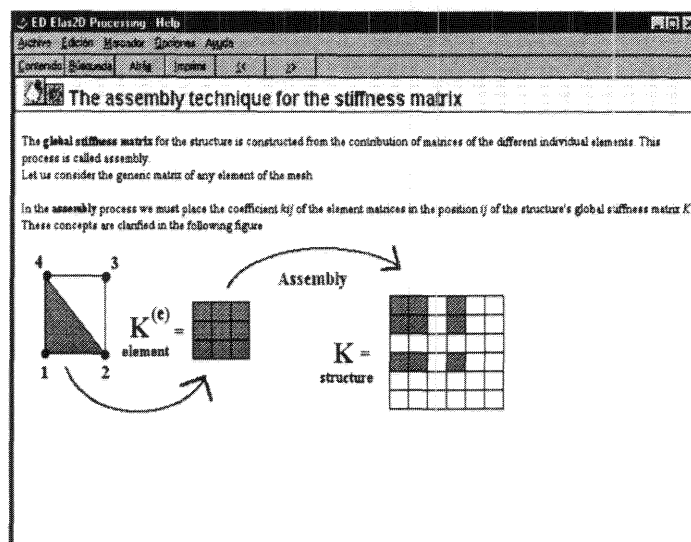


FIG. 7. Screen showing the theoretical lesson: stiffness matrix assembly.

- *Modularity*: that is, to allow for different learning and execution levels, ranging from a basic level for the beginner users to a more advanced level for those who are already familiar with FEM.
- *Individualization*: adjusted to the personal learning habit and adapted to the educational process to the user's environment.
- *Standardization*: that is, the development environment should not be completely new and should emulate other existing environments. In this manner, the effort of the users will be mainly directed to the acquisition of specific knowledge without investing much time learning the details on the use of the software environment.
- *User friendliness*: Indeed, the program must be easy to use in order that the learning process be as attractive to the user as possible.

All these considerations have been taken into account in the development of the educational environment ED-Elas2D. This educational program has been developed in the Windows environment and includes three distinct blocks which are a logical consequence of the basic steps to follow for the analysis of a structure with FEM. A brief description of these three blocks is given next.

2.1 Block 1: ED-Elas2D: Pre-processing

The pre-processing or data input block (Figs 1–4) allows one to introduce the necessary values defining the structure to be analyzed [6]. The geometry, the mechanical properties of the material, the generation of the mesh, the boundary conditions and the loading acting on the structure are defined here in an interactive manner. The pre-processing module has been designed so as to be very easy to use, with full graphic capacities and structured in a way so that the user can define all the data for the analysis of a specific structure in a very simple way. A number of options, such as drawing colours and fonts, allow one to visualize the definition of the structure. A help module is included in order to clarify any doubt which could arise during the edition of a data file for a specific problem.

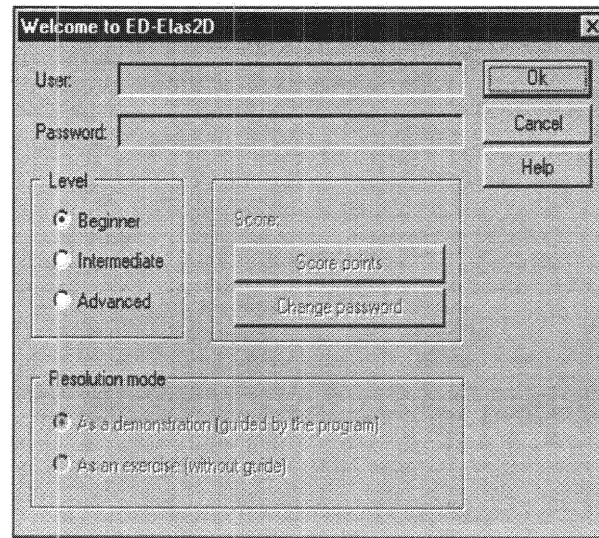


FIG. 8. Onset of the processing module of ED-Elas2D (levels and solution modes).

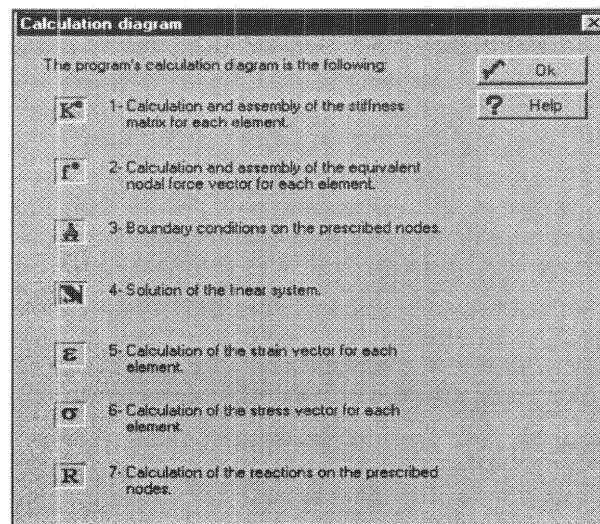


FIG. 9. Help menu in the beginner level showing the steps to solve a problem.

2.2 Block 2: ED-Elas2D: Processing

The solution process is the main block of the educational environment (Figs 5–15). This block includes the details for analysis of structures which obviously depend on the mathematical model chosen for the finite element solution. It also includes concepts such as the definition of the shape functions, the numerical integration, the assembly of the different terms of the stiffness matrix and the equivalent nodal load vector and also the solution of the linear system of equations.

From the pedagogical point of view, it is important that the user can be actively involved in all the solution steps which appear in the application of FEM. For this

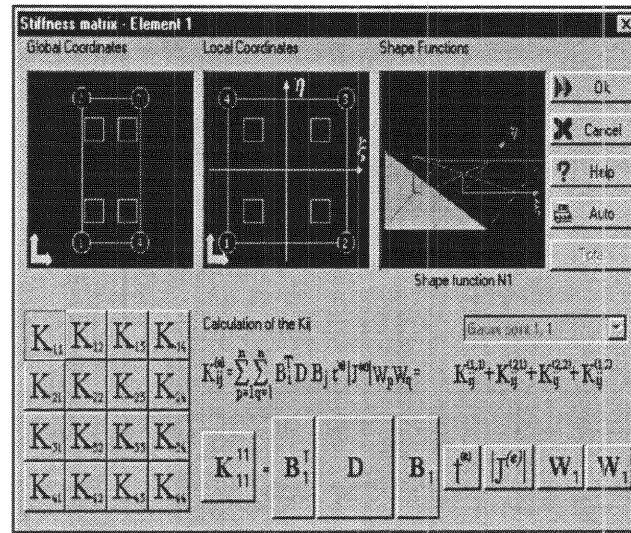


FIG. 10. Computation of the elements stiffness matrix. It shows the shape functions, the local/global numbering of nodes, etc.

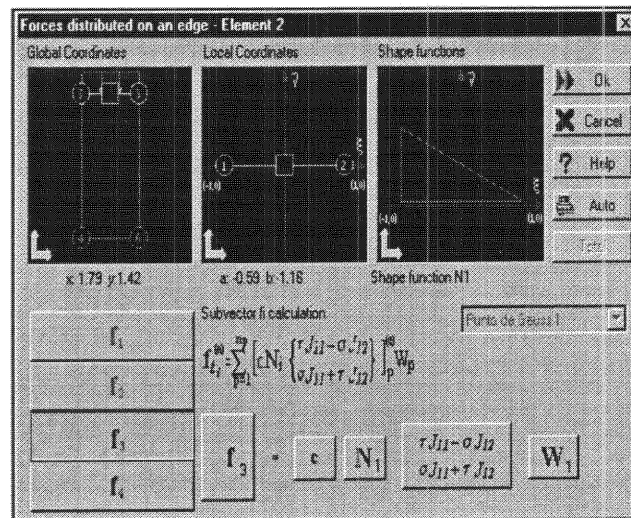


FIG. 11. Computation of the equivalent nodal force vector for an element. It shows the shape functions, the local/global numbering of nodes, etc.

purpose, ED-Elas2D includes a theory module based on the help of Windows [7], which facilitates any reference at any time during the solution process.

An important aspect which is worth pointing out is the facility to adapt the learning process to the specific knowledge of each student. For this purpose, the educational environment ED-Elas2D includes three solution levels to adjust the speed of the educational process to the degree of knowledge of each user [8].

The beginner level allows the user to follow the solution steps as if he/she were a mere observer while the computer controls the evolution of the process. At this level, only the example mode is allowed and the information flows in one unique direction according to

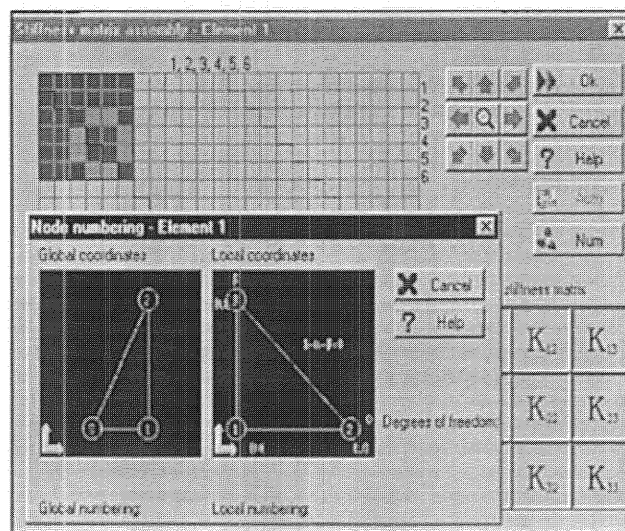


FIG. 12. Assembly of the stiffness matrix. It includes some utilities such as the auxiliary numbering of nodes.

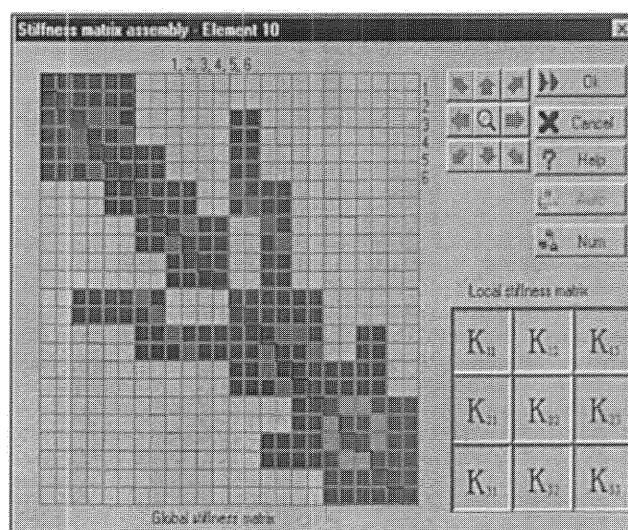


FIG. 13. Final assembly of the global stiffness matrix.

a deductive process. That is, once the process has been initiated the user can only move forward or exit. A higher flexibility is introduced by allowing the user to repeat some solution steps if required. The beginner level is recommended to users with none or little knowledge of FEM and therefore the problems which can be solved are limited to very simple cases.

The intermediate level assumes that the user has enough knowledge to take an active role in the solution of simple structures. This level allows two solution modes: examples and exercises. In the example mode the structures to be solved can be more complex than in the exercise mode. However, in this latter mode the solution process allows for a greater flexibility of options, that is the user can skip some steps, move backwards, try the solution

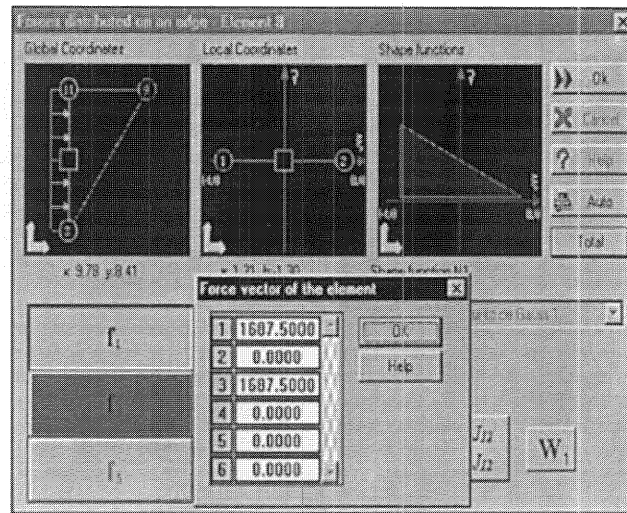


FIG. 14. Computation of the equivalent nodal force vector of an element. The numerical values can be displayed.

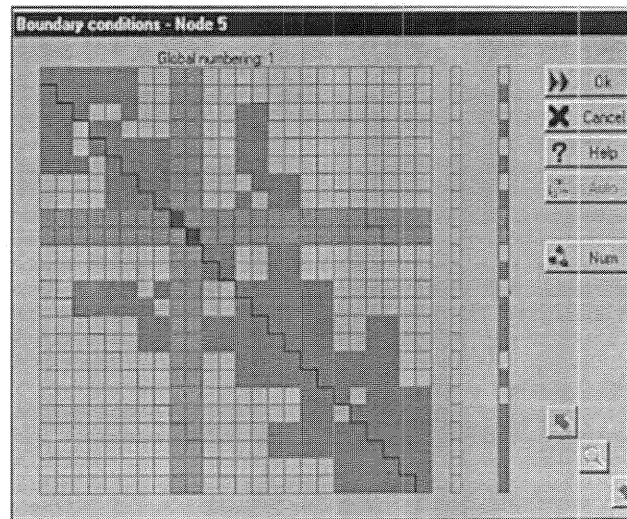


FIG. 15. Prescriptions of the boundary conditions.

of a partial step, etc. The learning process is here of inductive type and it makes use of the previous experience of the user.

The advanced level is addressed to users with a good knowledge of FEM and of the educational environment ED-Elas2D. Any type of exercise can be solved in this level, where the computer tutorial is minimum and the process is fully interactive. Now the user must guide the computer throughout the whole solution process, making use of all his/her knowledge while acquiring more practice in the use of FEM.

It has to be noted that the educational program ED-Elas2D has not been designed to perform a complete evaluation of the user's performance. However, the program

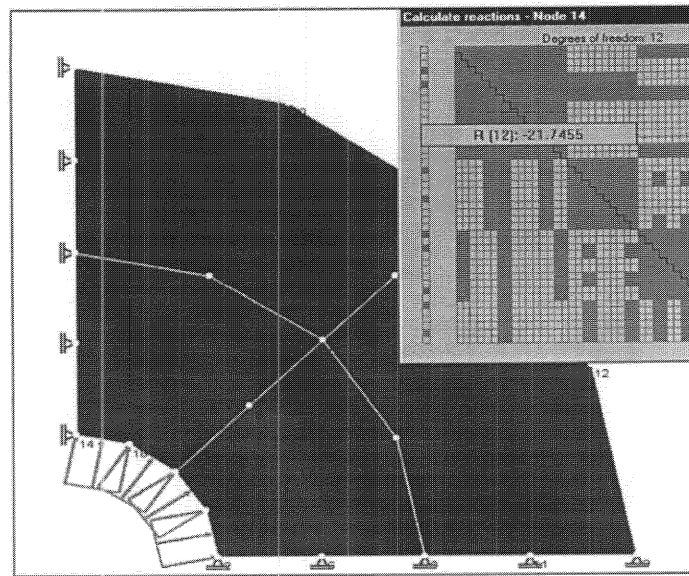


FIG. 16. Computation of reactions in a thick cylinder under internal pressure.

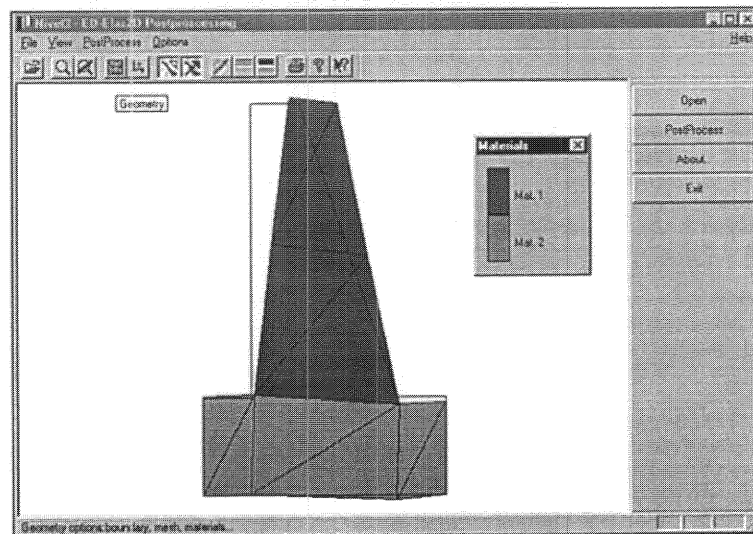


FIG. 17. ED-Elas2D. Post-processing. Geometry and deformed shape of a dam under self-weight and hydrostatic loading.

allows one to estimate the evolution of the learning process in quantitative form. Thus, once the solution of a problem is started the user can activate the marking option. Then marks will be automatically given by the computer according to the adequacy of the user's instructions. The total marks obtained by the user in the solution of a problem will allow him/her to access the next solution level, thus helping the user to control better his/her own progress.

2.3 Block 3: ED-Elas2D: Post-processing

The post-processing module (Figs 16–18) allows one to display the results of the

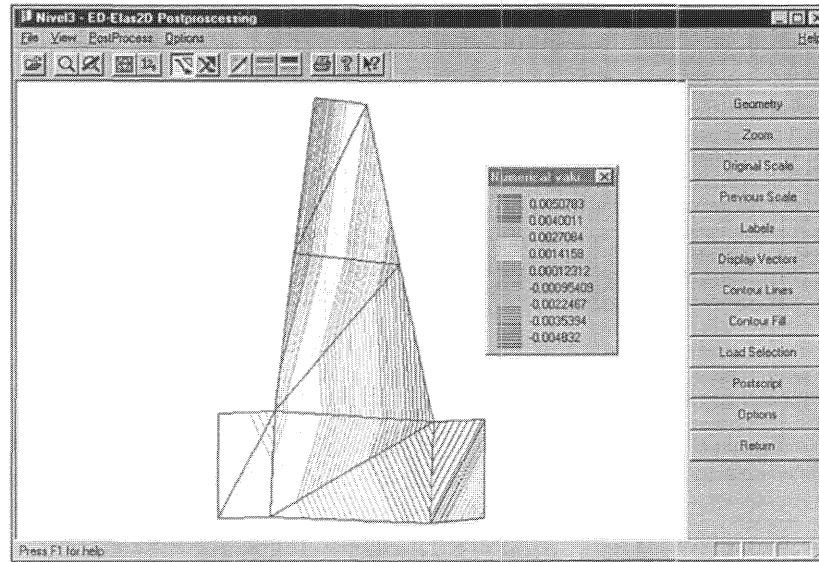


FIG. 18. Contours of the vertical displacement in a dam under self-weight and hydrostatic loading.

analysis in graphic form [6]. In this manner, the user can observe visually the main outputs of the finite element solutions (i.e. displacements, stress, strains, reactions, etc.). This module is developed following the standard of the Windows system [7] (cascade menus, button bar, tool bar, etc.).

The post-processing module, however, does not explain key concepts such as the nodal smoothing of stresses and strains. It is therefore a mere descriptive tool which simply provides visual information on the main results of the structural analysis.

3. Conclusions

The experience gained in the use of ED-Elas2D allows one to draw the following conclusions:

3.1 Work Environment

The Windows operating system guarantees an attractive graphic display and high interactivity between the computer and the user. The choice of a PC as the main working tool has proved to be adequate as this has allowed the programs to be accessible at a low cost. Use of the Microsoft standards also favours a quick learning of the program use.

3.2 Educational Environment

The organization of the program in three modular blocks, pre-processing, processing and post-processing, has also proved to be adequate as it follows the three logic steps in the practical solution of any structural problem by FEM. Each of the three blocks has its own educational content, which is of great importance for providing a sequential education into the different practical aspects of FEM.

The three learning levels (beginners, intermediate and advanced) allow the user to select the best learning strategy adequate to his/her knowledge. The flexibility of the

education reinforces the educational content of the program as the user can stop and repeat any solution steps in order to clarify or consolidate any idea.

The theory, example and exercises modes help to clarify the information to be transmitted to the user. The example mode has been designed to introduce the user into FEM methodology during the first steps of the learning process. Once the concepts have been assimilated the user can put them into practice by using the exercise mode.

A final comment on the evaluation of the user's progress. ED-Elas2D incorporates a simple marking scheme which quantifies the performance of the user in the solution of the structural problem selected. Further developments on more sophisticated marking procedures are in progress.

Acknowledgements

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REFERENCES

- [1] ED-ELAS2D (1996) *Educational Program for the Analysis of 2D Solids Using the Finite Element Method* (Barcelona, CIMNE).
- [2] ESCALANTE, F. (1991) *Una Metodología Para la Enseñanza Asistida por Ordenador en Ingeniería Estructural* (Barcelona, Escuela Técnica Superior de Ingenieros de Caminos, Canales y Puertos).
- [3] ED-BEAMS AND ED-FRAMES (1995) *Educational Programs for Structural Analysis Using Matrix Methods* (Barcelona, CIMNE).
- [4] SUÁREZ, B., GIL, L.L., TUGUÉS, R. & ONATE, E. (1994) *A Methodology for Computer-aided Education in Structural and Finite Analysis*, p. 489 (Paris, CALISCE).
- [5] ZIENKIEWICZ, O. & TAYLOR, R. (1991) *The Finite Element Method*, 4th Edn, Vol. II (New York, McGraw-Hill).
- [6] FEMGEN/FEMVIEW (1991) *The Finite Element Pre and Post-processor* (Leicester, Femview).
- [7] MICROSOFT CORPORATION (1990) *Microsoft Windows User's Guide for the Windows Graphical Environment*, Version 3.0.
- [8] BLANCO, E., GIL, L.L., SUÁREZ, B., ZAPATA, P. & ONATE, E. (1996) *Entorno Educativo ED-Elas2D, una Propuesta Para la Enseñanza del Método de los Elementos Finitos* (Zaragoza, SEMNI).